# AutoML, SHAP and Explanation Stability

From selecting a hypothesis function to explaining its predictions

## What do we select?
## What does the selected model explain?

A guided visual lecture with one customer prediction problem.

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# Our route through the problem

## Select and automate

30 minutes

## Explain with SHAP

30 minutes

## Connect

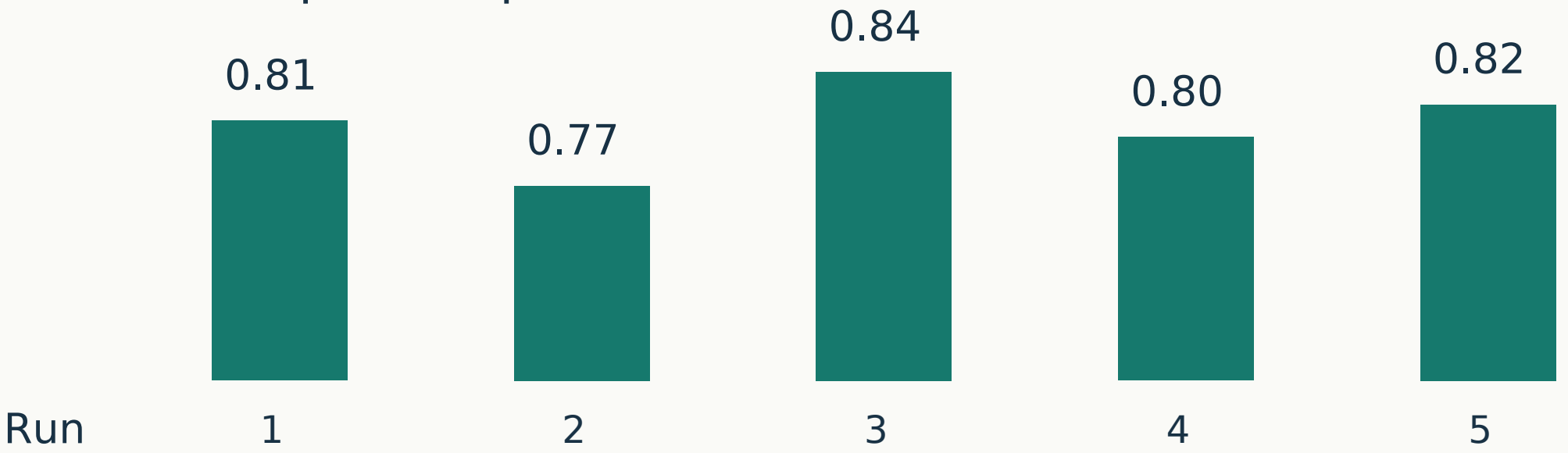
10 min

The slides guide the questions. The lab makes the choices and calculations visible.

# Last time: one run was one outcome

Illustrative repeated runs of the same experimental procedure

## The seed repeats a procedure. Several runs reveal variation.




Today: what procedure is choosing the model in the first place?

# The selection rule comes before the search

## Predict whether a customer will leave.

Eligible?
Mean CV recall ≥ 0.70

Among eligible candidates
Maximise mean CV precision

If no candidate meets the rule, report that. A search cannot invent an eligible solution.

# We select a recipe. Training learns a function.

Feature option
Recency only

→

Tree settings
Depth 4, leaf 15

→

Fitted tree
Splits and predictions

Chosen by the search

Learned from training data

The candidate is a pipeline configuration. A fitted hypothesis function depends on training data too.

# One candidate needs a fair evaluation




Fit within each fold, score on its validation portion, then average precision and recall separately.

# Lab 1: you are the selection algorithm

Open stage 1: Select a pipeline

| Features | Depth | Leaf |
|---|---|---|
| Both signals | 2 | 5 |
| Neither signal | 4 | 15 |
| Recency only | 4 | 15 |

Which candidate passes the rule? Which eligible candidate would you select?

# The search space is a set of possible recipes

4 feature options × 4 depths × 3 leaf sizes = 48 pipelines

| | Depth 2 | | | Depth 3 | | | Depth 4 | | | Depth 5 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Both signals | **0** | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
| Recency only | 12 | 13 | 14 | 15 | 16 | 17 | 18 | **19** | 20 | 21 | 22 | 23 |
| Inactivity only | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 |
| Neither signal | 36 | 37 | 38 | 39 | 40 | 41 | 42 | **43** | 44 | 45 | 46 | 47 |

Each depth has leaf sizes 5, 15 and 30. The highlighted squares are the three manual choices.

# AutoML automates parts of this selection process

We specify

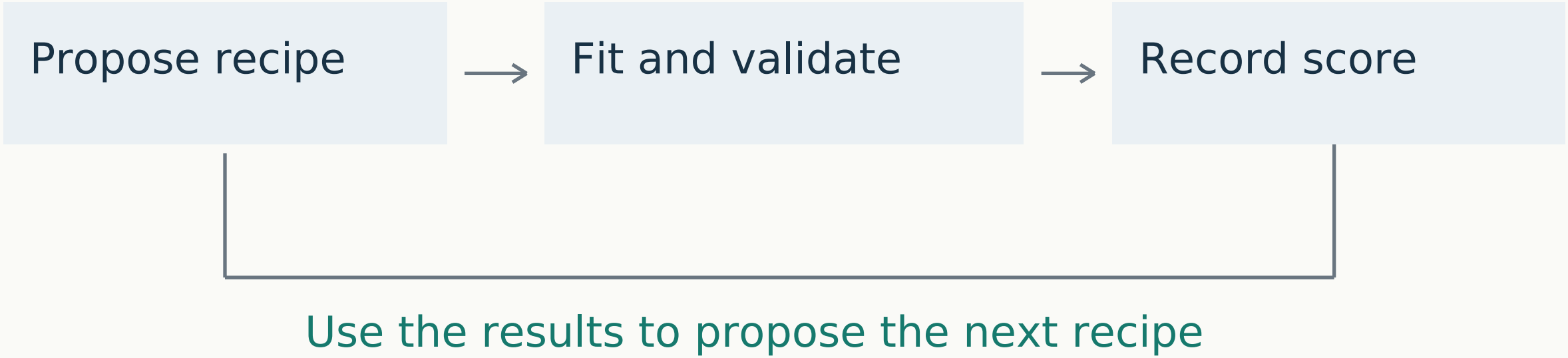
The task, available inputs, validation design, objective and budget.

The system searches

It proposes recipes, fits candidates and uses scores to decide what to try next.

AutoML can include preprocessing, feature choices, model families and hyperparameters.

# The search loop surrounds model fitting




Stop at the declared budget. Keep the best eligible candidate found.

# Frameworks package the machinery

Examples of different design choices

| Framework | Search idea | What it helps organise |
| --- | --- | --- |
| auto-sklearn | Bayesian optimisation | Pipeline search and ensembles |
| TPOT | Evolutionary search | Pipeline construction and evaluation |

A framework supplies an implementation. Its defaults still express modelling choices.

# Automation inherits our decisions and mistakes

| If we supply... | The consequence can be... |
| --- | --- |
| A poor metric | A strong score for the wrong goal |
| An incomplete search space | No access to a useful candidate |
| Leaked or unsuitable validation | Efficient selection of misleading results |
| A small budget | A result shaped by the visited candidates |

# Why consider evolutionary search?

The choices are mixed

Feature subsets, model families and integer settings do not form one smooth surface.

A population explores alternatives

Selection uses validation feedback. Variation creates new recipes.

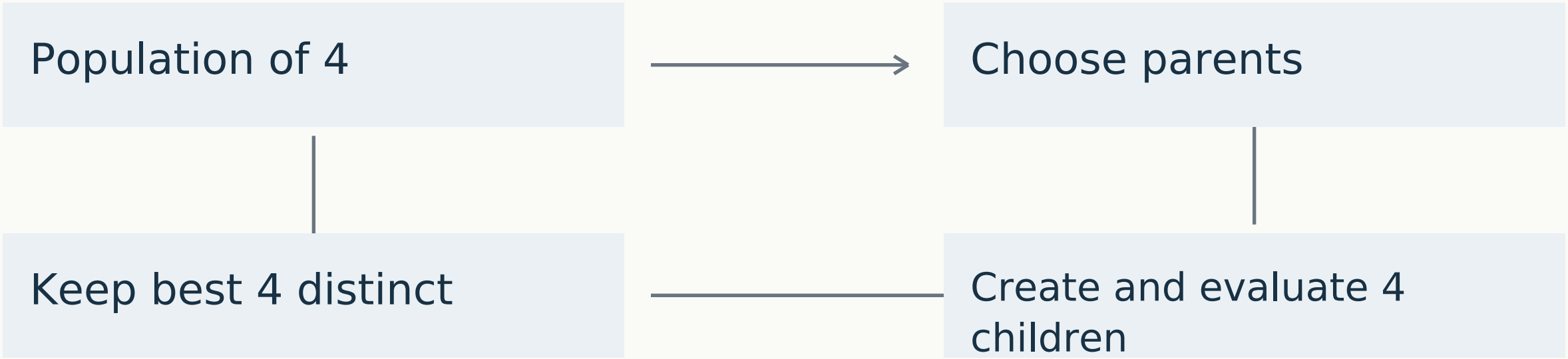
Evolution still costs evaluations, can converge early and needs comparison with simple baselines.

# A candidate can be encoded as a chromosome

| Feature choice | Maximum depth | Minimum leaf size |
| --- | --- | --- |
| Recency only | 4 | 15 |

Changing a gene changes the recipe. We must train and evaluate the new candidate.

# Selection, variation, evaluation, survival




A population is a set of candidate pipelines. A generation is one round of this loop.

# Recombination and mutation change the recipe

| Candidate | Features | Depth | Leaf |
|---|---|---|---|
| Parent A | Recency only | 4 | 15 |
| Parent B | Both signals | 2 | 30 |
| Recombined child | Recency only | 2 | 30 |
| After mutation | Recency only | 5 | 30 |

# Lab 2: one generation, slowly

Open stage 2: Automate the search

## Before clicking: what needs to happen to score a child?

### Watch the recipes

Which choices came from each parent? Which choice changed?

### Watch the evidence

Did the child pass recall? Did it improve the best eligible precision?

# A search result depends on its budget and seed

## Hold fixed

Candidate space, folds, scoring rule and evaluation budget.

## Repeat the search

Change search seeds. Record outcomes, failures and evaluation costs.

This 48-candidate demo explains the mechanism. It cannot establish that evolution is superior.

# A good score leaves another question

Why did this fitted model make this prediction?

Customer 1

Selected model → Leave: 90.8%

An explanation describes how the model uses information to produce an output.

# Different questions need different explanations

| Question | Scope |
| --- | --- |
| Why this prediction for customer 1? | Local explanation |
| Which inputs receive large attributions across customers? | Global summary |
| Would changing customer support prevent leaving? | Causal question |

# A prediction needs a reference for comparison

**Reference prediction**
**46.8%**

Average output on 60 background customers

**Customer 1 prediction**
**90.8%**

Output for this particular input

Which input features receive credit for the difference?

# SHAP allocates a prediction difference

## SHapley Additive exPlanations

Reference output + feature contributions = model output

Each contribution averages what a feature adds when it joins different groups of other features.

# What does an absent feature mean?

Keep customer values
for present features

Use background values
for absent features

## Predict for each completed input, then average the outputs.

The fitted model stays fixed. The background and replacement rule define the explanation question.

# Two features let us calculate the idea by hand

Illustrative coalition values, independent of the larger lab model

| Features revealed | Average predicted probability |
| --- | --- |
| Neither | 30% |
| A only | 50% |
| B only | 40% |
| A and B | 70% |

# Reveal A first, then B

Neither
30%

→

A only
50%

→

A and B
70%

A adds 20 points

B adds 20 points

# Reveal B first, then A

Neither
30%

→

B only
40%

→

A and B
70%

B adds 10 points

A adds 30 points

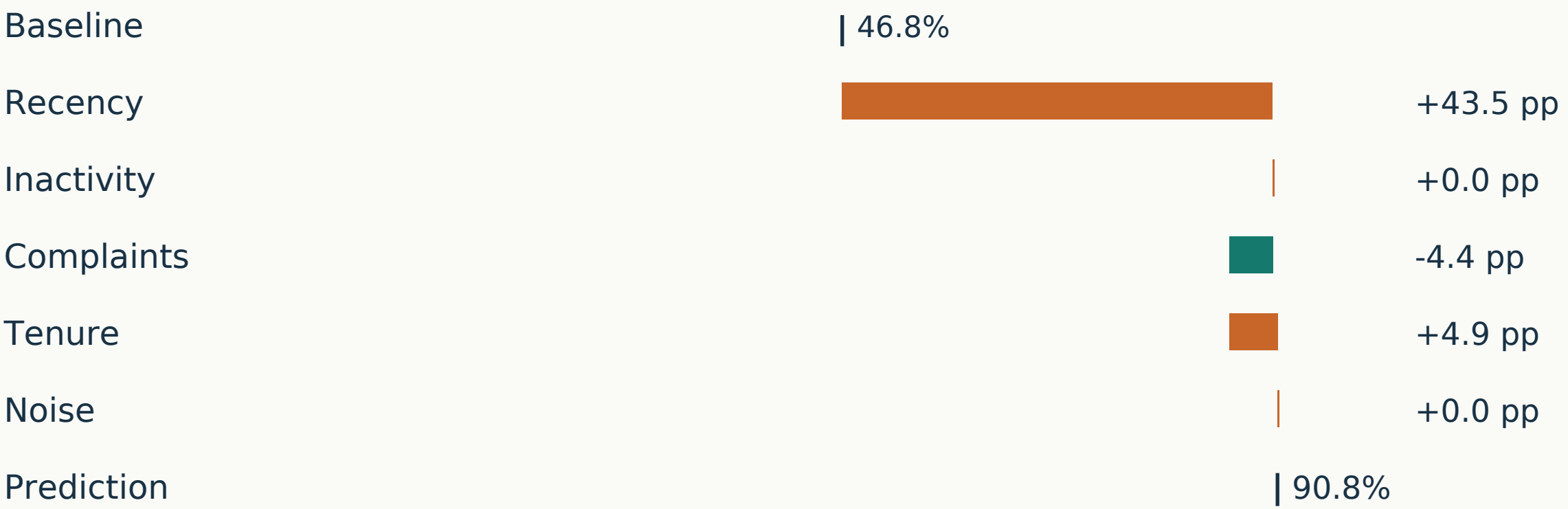
# Shapley values average across the orders

| Feature | Marginal contributions | Average |
|---|---|---|
| A | 20 and 30 points | +25 points |
| B | 20 and 10 points | +15 points |

30% + 25 points + 15 points = 70%

# A waterfall connects the baseline to the prediction

Actual exact Shapley values for M19 and customer 1



# A negative contribution lowers this model output

## Positive SHAP value

Pushes this prediction above the reference output.

## Negative SHAP value

Pushes this prediction below the reference output.

A contribution sign does not tell us the effect of intervening on a real customer.

# Lab 3: build one explanation, one bar at a time

Open stage 3: Explain with SHAP

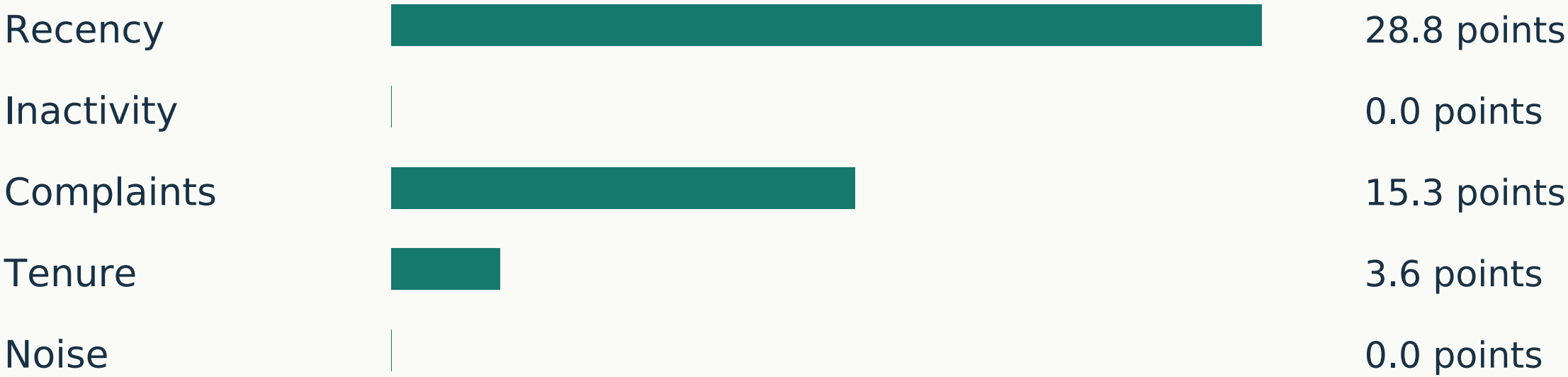
Hold the model fixed. Choose customer 1.

Read the baseline, then reveal each feature contribution.

Change the customer. Which parts of the explanation change?

# Local explanations can be summarised globally

Same model, same background, 100 audit customers




Mean absolute SHAP measures typical attribution magnitude. Taking absolute values discards direction.

# A SHAP value needs its context

| Record | In this lab |
| --- | --- |
| Fitted model and output | M19, probability of leaving |
| Case or sample | Customer 1, or 100 audit customers |
| Background | 60 fixed development customers |
| Absent-feature rule | Empirical interventional replacement |

# A precise explanation still has limits

## Inputs can overlap

Replacing one correlated input may create unusual combinations.

## The question can change

A different background or dependence assumption can change the attribution.

SHAP explains a model output under stated assumptions. It does not prove causality or model validity.

# An AI partner can help us check a claim

“AutoML found the best model. SHAP proves that inactivity causes customers to leave.”

What evidence is missing from each sentence?

Ask the AI partner for a counterexample, then check it against the search and explanation ledger.

# What we can now explain

The search selects a pipeline configuration using validation evidence.

SHAP allocates a fixed model output relative to a declared reference.

Now a research question: would another similarly performing model tell the same story?

# Research glimpse: one good model may not be unique

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|---|---|---|---|---|---|---|---|---|---|---|---|
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Several candidate pipelines can fit nearly as well under the same evaluation rule.

# The Rashomon idea keeps near-optimal alternatives

Keep eligible candidates within 2 precision points of the best tested eligible candidate.

| Best observed | Cutoff | In the set |
| --- | --- | --- |
| 83.29% | 81.29% | 9 of 48 |

# Similar scores can support different attributions

## M19: Recency only

Precision 83.25%
Prediction 90.8%

Recency: +43.5 points

Inactivity: +0.0 points

## M31: Inactivity only

Precision 83.25%
Prediction 90.8%

Recency: +0.0 points

Inactivity: +43.5 points

Lab 4: same customer, same background, same SHAP definition. Only the fitted model changes.

# Stability requires saying what varies

| Change | Keep fixed to isolate it |
| --- | --- |
| Training sample or seed | Recipe, audit cases, explanation settings |
| Selected model | Audit cases, background, SHAP definition |
| Background sample | Fitted model and cases |

Persistent attribution is evidence of robustness to that change. It does not establish causal truth.

# The whole experiment needs an honest claim

We searched a declared space under a fixed budget.

We explained a specified output with a specified reference.

We checked whether the explanation persisted across alternatives.

Your research connection: dependable predictions and dependable explanations are related questions.

# Sources for AutoML and explanation


Olson and Moore (2016). TPOT: A Tree-based Pipeline Optimization Tool.

https://proceedings.mlr.press/v64/olson_tpot_2016.html

Lundberg and Lee (2017). A Unified Approach to Interpreting Model Predictions.

https://proceedings.neurips.cc/paper_files/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html

Fisher, Rudin and Dominici (2019). All Models are Wrong, but Many are Useful.

https://jmlr.org/papers/v20/18-760.html

# Lab methods and further reading

Synthetic independent customers. Shared 3-fold CV on 900 development cases. Fixed tree seed. A separate audit set supports exploratory explanations. The reserved test set stays unused.

Exact Shapley values enumerate 32 feature subsets using 60 fixed background rows. Local additivity is checked for every model and audit case.

Framework documentation: automl.github.io/auto-sklearn/master/manual.html
TPOT: epistasislab.github.io/tpot/latest/
SHAP: shap.readthedocs.io/en/latest/generated/shap.TreeExplainer.html